

GIT - COLLABORATE IN PROJECT USING GITHUB

❖ GitHub Dashboard Overview

- The GitHub Dashboard is your control panel to **manage repositories, contributions, and interactions** with other developers.

❖ Difference Public vs Private Repositories

Feature	Public Repository	Private Repository
Visibility	Everyone	Only collaborators
Security	Lower (open access)	Higher (restricted access)
Use Case	Open-source, education	Enterprise, confidential work
Cost (GitHub Free)	Free	Free (limited private repos)
Forking	Anyone can fork	Not allowed unless made public

❖ GitHub Explore Page:

Go to

<https://github.com/explore>

❖ Forks and Pull Requests (PRs)

▪ What is Forking?

- Forking creates a personal copy of someone else's repository on your GitHub account.

▪ Use Cases:

- Contribute to open-source projects
- Experiment with code without affecting the original
- Create your own version of a project

▪ How to Fork:

- Open the repository on GitHub.
- Click the **"Fork"** button (top-right).
- GitHub will copy it to your account

▪ What is a Pull Request?

- A pull request proposes changes from your fork/branch into the original repository.

▪ Steps:

1. Fork a repository
2. Make changes (add commits) to your fork
3. Click "Compare & pull request"
4. Submit the PR with a message describing your changes
5. The repository owner reviews and merges if appropriate

▪ Pull Request Life Cycle:

- Fork → Edit → Commit → Pull Request → Review → Merge

❖ If prompted, use:

- **Username:** your GitHub username
- **Password:** your Personal Access Token
 - Create a PAT at: [🔗](#)
 - settings/tokens
 - github.com/settings/tokens

❖ SSH Authentication

- **Use Case:** Best for long-term contributors, secure and password-less access.
- **Cons:**
 - Requires SSH key setup
 - May be blocked in strict firewalls
- **Steps: #**
 - Generate SSH key
 - `ssh-keygen -t ed25519 -C "your_email@example.com"` Add to `ssh-agent`
`eval "$(ssh-agent -s)"`
 - `ssh-add ~/.ssh/id_ed25519`
 - # Add public key to GitHubcat
~/.ssh/id_ed25519.pub
 - Paste the public key into GitHub:
 - [🔗](#) GitHub → Settings → SSH and GPG Keys → New SSH Key

[General Interview Questions]

Q1. What is the difference between a public and a private repository?

- A **public repository** is visible to everyone and can be viewed or forked by anyone.
- A **private repository** is restricted to specific collaborators and is not publicly accessible.

Q2. Can you convert a public repository to private on GitHub?

- **Yes**, GitHub allows you to change the visibility of a repository from public to private and vice versa via the repository settings.

Q3. What does “forking” a repository mean?

- Forking creates a copy of someone else’s repository under your own GitHub account, allowing you to freely experiment without affecting the original project.

Q4. What is a pull request in GitHub?

- A pull request is a way to propose changes to a repository by comparing changes from your fork or branch to the original project.
- It’s commonly used for code review and collaboration.

Q5. Can you fork a private repository?

- **No**, private repositories cannot be forked unless the owner explicitly grants permission and both repos remain private.

Q6. What are the common authentication methods in GitHub?

- The two most common authentication methods are **HTTPS** (with username and token) and **SSH keys**.

Q7. What is the advantage of using SSH over HTTPS for Git operations?

- SSH allows passwordless and more secure access after initial setup, while HTTPS requires credential input or token each time unless cached.